ML MCQ Practice Set

InMobi / Glance — Applied Scientist I online assessment

100 questions, each with one correct answer, grouped by the topics the job description lists: ML fundamentals, trees and clustering, neural networks, transformers, evaluation methodology, statistics, bandits and RL, recommendation and ads, and the Python data stack. Allow about 90 seconds per question. The answer key with explanations starts on page 19.

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Supervised Learning Fundamentals

Q1. Increasing k in a k-nearest-neighbours classifier typically:

(a) Decreases both bias and variance

(b) Has no effect on the bias–variance trade-off

(c) Increases variance and decreases bias

(d) Increases bias and decreases variance

Q2. Why does L1 regularisation (Lasso) tend to produce sparse weight vectors while L2 (Ridge) does not?

(a) The L1 constraint region has corners on the coordinate axes, where the loss contours tend to touch it

(b) L1 corresponds to a Gaussian prior on the weights

(c) L1 penalises large weights more heavily than L2

(d) The L1 penalty is differentiable everywhere

Q3. Ridge regression is equivalent to MAP estimation of linear-regression weights under which prior?

(a) Zero-mean Gaussian

(b) Zero-mean Laplace

(c) Uniform

(d) Beta(1,1)

Q4. Training logistic regression by minimising binary cross-entropy is equivalent to:

(a) Minimising the hinge loss

(b) Maximising the geometric margin between classes

(c) Maximum-likelihood estimation of a Bernoulli model with $p = \sigma(w^\top x)$

(d) Minimising squared error between predicted probabilities and labels

Q5. A model has 2% training error and 25% validation error. Which action is *least* likely to help?

(a) Increasing regularisation strength

(b) Early stopping

(c) Collecting more training data

(d) Adding many more polynomial feature interactions

Q6. The normal-equation solution $w = (X^\top X)^{-1} X^\top y$ fails because $X^\top X$ is singular when:

(a) The features are not standardised

(b) The target y is not normally distributed

(c) Some features are perfectly collinear, or there are more features than samples

(d) The dataset has more samples than features

Q7. In a trained soft-margin SVM, which training points determine the decision boundary?

(a) Only the misclassified points

(b) All training points equally

(c) Only the support vectors: points on or inside the margin, or misclassified

(d) Only the points farthest from the boundary

Q8. The RBF (Gaussian) kernel $k(x, z) = \exp(-\gamma \|x - z\|^2)$ corresponds to an inner product in a feature space of dimension:

(a) Infinite

(b) Equal to the number of training points

(c) Equal to the input dimension d

(d) d^2

Q9. The "naive" assumption in Naive Bayes is that:

(a) Features are marginally independent

(b) Features are conditionally independent given the class

(c) Classes are equally likely a priori

(d) Every feature is Gaussian

Q10. Which of these is a *generative* classifier?

(a) A feed-forward neural network with softmax output

(b) Logistic regression

(c) Gaussian Naive Bayes

(d) Linear SVM

Q11. For which model does standardising (scaling) the features matter *least*?

(a) L2-regularised logistic regression

(b) A decision tree

(c) k-nearest neighbours

(d) An SVM with an RBF kernel

Q12. For squared-error loss, the expected test error at a point decomposes as:

(a) Variance + Noise only, since bias cancels in expectation

(b) Bias + Variance

(c) Bias^2 + Variance + Irreducible noise

(d) $\text{Bias}^2 -$ Variance + Noise

Decision Trees and Ensembles

Q13. A node contains 50% positives and 50% negatives. Its Gini impurity is:

(a) 0.5

(b) 1

(c) 0

(d) 0.25

Q14. A parent node has entropy 1.0 bit. A split sends half the samples to a pure child (entropy 0) and half to a child with entropy 0.918. The information gain is approximately:

(a) 0.918

(b) 0.459

(c) 0.082

(d) 0.541

Q15. The main reason a random forest generalises better than a single deep tree is that it:

(a) Uses a different loss function per tree

(b) Prunes every tree to depth 1

(c) Reduces bias by fitting each tree to the previous trees' residuals

(d) Reduces variance by averaging many decorrelated trees (bootstrap samples + random feature subsets)

Q16. In gradient boosting, each new tree is fit to:

(a) The negative gradient of the loss with respect to the current ensemble's predictions

(b) A bootstrap sample of the original labels

(c) The original labels with misclassified points reweighted

(d) Random noise, to decorrelate the ensemble

Q17. When you lower the learning rate (shrinkage) in gradient boosting, you usually need to:

(a) Increase the number of trees

(b) Decrease the number of trees

(c) Increase tree depth to infinity

(d) Remove the regularisation terms

Q18. Approximately what fraction of the training points are left out of a single bootstrap sample of size n (for large n)?

(a) 50%

(b) 10%

(c) 63.2%

(d) 36.8%

Q19. After each round, AdaBoost:

(a) Resamples uniformly at random

(b) Discards the misclassified samples

(c) Increases the weights of misclassified samples

(d) Decreases the weights of misclassified samples

Q20. A single fully grown (unpruned) decision tree typically has:

(a) High bias and high variance

(b) Low bias and high variance

(c) High bias and low variance

(d) Low bias and low variance

Q21. XGBoost's split-gain formula uses which statistics of the loss?

(a) Only the first-order gradient

(b) Only the Gini impurity of each child

(c) First- and second-order derivatives (gradient and Hessian)

(d) The raw label counts in each child

Q22. Impurity-based (mean decrease in impurity) feature importance in random forests is known to be biased towards:

(a) High-cardinality or continuous features

(b) Binary features

(c) Features with missing values

(d) Features that appear late in the column order

Unsupervised Learning and Clustering

Q23. The objective that k-means minimises is:

(a) The within-cluster sum of squared Euclidean distances to the centroids

(b) The maximum distance between any two points in a cluster

(c) The number of points assigned to each cluster

(d) The negative log-likelihood of a Gaussian mixture with full covariances

Q24. On which data does k-means usually perform worst?

(a) Two concentric rings

(b) Well-separated spherical blobs of equal size

(c) Data whose clusters are linearly separable and convex

(d) Blobs with equal variance and equal counts

Q25. k-means++ initialisation chooses each new centre:

(a) As the point farthest from all chosen centres, deterministically

(b) With probability proportional to its squared distance from the nearest centre already chosen

(c) Uniformly at random from the data

(d) As the mean of a random subset of points

Q26. Which is a property of DBSCAN?

(a) It works best when clusters have very different densities

(b) It can find arbitrarily shaped clusters and labels low-density points as noise

(c) It requires the number of clusters as input

(d) It always assigns every point to some cluster

Q27. PCA on a dataset gives eigenvalues 6, 3 and 1. What fraction of the variance do the first two components explain?

(a) 67%

(b) 60%

(c) 100%

(d) 90%

Q28. The silhouette score of a clustering lies in which range, and what does a value near $+1$ indicate?

(a) $[0, \infty)$; very low inertia

(b) $[-1, 1]$; points are much closer to their own cluster than to the nearest other cluster

(c) $[0, 1]$; clusters have equal size

(d) $[-1, 1]$; points lie on the boundary between clusters

Q29. Compared with k-means, a Gaussian mixture model fitted with EM:

(a) Gives soft (probabilistic) assignments and can model elliptical clusters

(b) Always finds the global optimum

(c) Can only model spherical clusters of equal size

(d) Needs no choice of the number of components

Q30. Agglomerative clustering with *single* linkage is prone to:

(a) Chaining: merging distinct clusters connected by a thin path of points

(b) Producing only clusters of equal size

(c) Ignoring outliers entirely

(d) Requiring a Gaussian assumption

Neural Networks and Optimisation

Q31. Why do deep networks with sigmoid activations suffer from vanishing gradients?

(a) The sigmoid is not differentiable at 0

(b) The sigmoid's derivative is at most 0.25, so backpropagated gradients shrink multiplicatively through layers

(c) The sigmoid output can be negative

(d) The sigmoid derivative grows without bound for large inputs

Q32. A "dead" ReLU unit is one that:

(a) Outputs 0 for all inputs, so it receives zero gradient and stops learning

(b) Always outputs its input unchanged

(c) Outputs very large values and causes exploding gradients

(d) Has a learning rate of 0

Q33. At inference time, a batch-normalisation layer normalises with:

(a) Statistics recomputed from the whole training set at each call

(b) Mean 0 and variance 1 regardless of data

(c) The mean and variance of the current inference batch

(d) Running averages of mean and variance accumulated during training

Q34. With inverted dropout and drop probability $p = 0.5$, what happens to activations?

(a) Activations are multiplied by 0.5 in both training and test

(b) During training nothing is scaled; at test time activations are multiplied by 0.5

(c) During training, kept activations are multiplied by 2; at test time no scaling is applied

(d) Dropout is also applied at test time, with the same p

Q35. Adam combines which two ideas?

(a) Dropout and weight decay

(b) Newton's method and line search

(c) Momentum (first moment of gradients) and per-parameter scaling by the second moment, as in RMSProp

(d) Batch normalisation and gradient clipping

Q36. For a deep network with ReLU activations, the recommended weight initialisation is:

(a) Glorot initialisation, $\mathrm{Var}(w) = 2/(\text{fan_in} + \text{fan_out})$, which was derived for ReLU

(b) He initialisation, $\mathrm{Var}(w) = 2/\text{fan_in}$

(c) All weights set to zero

(d) Uniform on $[-10, 10]$

Q37. How many trainable parameters does a fully connected layer from 128 inputs to 64 outputs (with bias) have?

(a) 8,256

(b) 192

(c) 8,320

(d) 8,192

Q38. A conv layer has 16 input channels, 32 filters of size 3×3, and biases. How many parameters does it have?

(a) 320

(b) 4,640

(c) 9,248

(d) 4,608

Q39. A 32×32 input passes through a conv layer with a 3×3 kernel, stride 2, padding 1. The output's spatial size is:

(a) 30×30

(b) 15×15

(c) 32×32

(d) 16×16

Q40. For a softmax output p with cross-entropy loss and one-hot label y, the gradient of the loss with respect to the logits z is:

(a) $y - p$

(b) $-\log p_y$

(c) $p(1-p)$

(d) $p - y$

Q41. Training loss oscillates wildly and then becomes NaN in the first few hundred steps. The most likely cause is:

(a) Too much L2 regularisation

(b) The validation set is too small

(c) The learning rate is too high

(d) The model has too few parameters

Q42. How does AdamW differ from Adam with an L2 penalty added to the loss?

(a) Weight decay is applied directly to the weights, decoupled from the adaptive gradient scaling

(b) AdamW does not use momentum

(c) There is no difference; the two are identical

(d) AdamW uses a second-order Hessian

Transformers

Q43. Scaled dot-product attention divides $QK^\top$ by $\sqrt{d_k}$ because:

(a) It normalises each row to sum to 1

(b) It reduces the computational complexity

(c) The variance of dot products grows with d_k, pushing the softmax into saturated regions with tiny gradients

(d) It makes the attention matrix symmetric

Q44. The time complexity of self-attention in sequence length n (model width d) is:

(a) $O(nd^2)$ only

(b) $O(n^2 d)$

(c) $O(n \log n)$

(d) $O(n)$

Q45. The causal mask in a decoder-only transformer:

(a) Randomly masks 15% of the input tokens

(b) Prevents each position from attending to later positions

(c) Prevents attention to the first token

(d) Removes padding tokens from the vocabulary

Q46. The main benefit of multi-head attention over a single head of the same total width is that it:

(a) Makes attention linear in sequence length

(b) Reduces the parameter count by a factor of h

(c) Lets the model attend to information from different representation subspaces at the same time

(d) Removes the need for positional information

Q47. Transformers need positional encodings because:

(a) Layer normalisation removes position information

(b) Embeddings cannot be learned otherwise

(c) Self-attention is permutation-equivariant, so it has no notion of token order on its own

(d) Softmax requires positive inputs

Q48. BERT is pre-trained mainly with:

(a) Contrastive image–text matching

(b) Masked language modelling (predicting randomly masked tokens using both left and right context)

(c) Next-token prediction with a causal mask

(d) Reinforcement learning from human feedback

Q49. During autoregressive generation, the KV cache:

(a) Caches the gradients for backpropagation

(b) Stores the keys and values of previous tokens so they are not recomputed at each new step

(c) Compresses the vocabulary embedding table

(d) Stores the final output probabilities

Q50. Transformers use layer normalisation rather than batch normalisation mainly because layer norm:

(a) Has no learnable parameters

(b) Is only applied at inference time

(c) Normalises across the batch for each feature

(d) Normalises across features within each token, independent of batch size and sequence padding

Q51. Ignoring biases, how many parameters are in the Q, K, V and output projections of one multi-head attention layer with $d_{\text{model}} = 512$?

(a) 2,097,152

(b) 786,432

(c) 262,144

(d) 1,048,576

Q52. Raising the sampling temperature T above 1 when generating from a language model:

(a) Has no effect on the output distribution

(b) Sharpens the distribution towards the most likely token

(c) Restricts sampling to the top-k tokens

(d) Flattens the token distribution, making outputs more diverse and random

Evaluation Methodology and Metrics

Q53. A classifier has TP $= 40$, FP $= 10$, FN $= 20$. Its F1 score is approximately:

(a) 0.73

(b) 0.80

(c) 0.57

(d) 0.67

Q54. ROC-AUC equals the probability that:

(a) The model's precision exceeds its recall

(b) A randomly chosen positive is scored higher than a randomly chosen negative

(c) A randomly chosen prediction is correct

(d) The predicted probability equals the true probability

Q55. For an ad click model where only 0.1% of impressions are clicks, which metric is usually more informative for comparing models?

(a) ROC-AUC, because it is insensitive to class balance

(b) PR-AUC (average precision)

(c) Accuracy

(d) Specificity

Q56. Which of these is an example of data leakage?

(a) Using a fixed random seed for the split

(b) Fitting a StandardScaler on the full dataset before splitting into train and test

(c) Tuning hyperparameters on a validation set separate from the test set

(d) Using stratified k-fold cross-validation

Q57. You build a CTR model from six months of ad logs. The most appropriate validation scheme is:

(a) Train on earlier days and validate on later days (time-based split)

(b) Random 80/20 split of all impressions

(c) Leave-one-out cross-validation

(d) Validate on the training data with bootstrapping

Q58. Which metric is unchanged if you apply a strictly increasing transformation to all predicted scores?

(a) Log loss

(b) ROC-AUC

(c) Brier score

(d) Expected calibration error

Q59. A ranked list has binary relevances $[0, 1, 1]$ at positions 1, 2, 3. Using $\text{DCG} = \sum_i \text{rel}_i / \log_2(i+1)$, NDCG@3 is approximately:

(a) 1.13

(b) 0.50

(c) 0.75

(d) 0.69

Q60. Across three queries, the first relevant result appears at ranks 1, 2 and 4. The mean reciprocal rank is:

(a) 0.333

(b) 0.750

(c) 2.333

(d) 0.583

Q61. You want both to tune hyperparameters and to report an unbiased estimate of generalisation performance from a small dataset. You should use:

(a) A single k-fold CV, reporting the best fold score

(b) Leave-one-out CV with the tuned hyperparameters chosen on the same folds

(c) Training-set accuracy

(d) Nested cross-validation

Q62. Stratified k-fold cross-validation ensures that:

(a) Each fold contains data from only one time period

(b) Each fold has roughly the same class proportions as the full dataset

(c) Each user's data appears in only one fold

(d) The folds have different sizes

Q63. A new ranking model improves offline AUC, but an online A/B test shows no CTR gain. Which is a likely explanation?

(a) AUC always perfectly predicts online CTR

(b) Offline data is biased by the old model's exposure and position (feedback loop / training–serving mismatch)

(c) The A/B test had too large a sample

(d) The model's learning rate was too low

Q64. Two classifiers are evaluated on the *same* test set. To test whether their error rates differ significantly, a suitable test is:

(a) The Kolmogorov–Smirnov test on the features

(b) McNemar's test

(c) A chi-squared goodness-of-fit test on the label distribution

(d) An independent two-sample t-test on their accuracies

Probability and Statistics

Q65. A disease has 1% prevalence. A test has 99% sensitivity and a 5% false-positive rate. Given a positive test, the probability of disease is about:

(a) 50%

(b) 95%

(c) 17%

(d) 99%

Q66. A p-value is:

(a) The probability, assuming H_0 is true, of observing data at least as extreme as what was observed

(b) The probability that H_1 is true

(c) The probability that H_0 is true

(d) The probability of making a Type II error

Q67. In an A/B test on CTR, you increase the sample size per arm by a factor of 4. The standard error of the difference in CTR:

(a) Stays the same

(b) Quarters

(c) Halves

(d) Doubles

Q68. The statistical power of a test is:

(a) $P(H_0 \text{ true} \mid \text{data})$

(b) $P(\text{reject } H_0 \mid H_0 \text{ true})$

(c) $P(\text{reject } H_0 \mid H_1 \text{ true})$

(d) $P(\text{fail to reject } H_0 \mid H_1 \text{ true})$

Q69. You test 20 independent metrics in an experiment, each at $\alpha = 0.05$, and no true effect exists. The probability of at least one false positive is about:

(a) 64%

(b) 20%

(c) 100%

(d) 5%

Q70. A bootstrap confidence interval is built by:

(a) Resampling the observed data with replacement many times and taking quantiles of the recomputed statistic

(b) Assuming the statistic is normally distributed and using ± 1.96 SE only

(c) Splitting the data into k folds without replacement

(d) Collecting new data from the population for each replicate

Q71. An ad gets 7 clicks in 100 impressions. With a Beta(1,1) prior on CTR, the posterior mean CTR is:

(a) $7/102 \approx 0.069$

(b) $7/100 = 0.070$

(c) $8/102 \approx 0.078$

(d) $8/100 = 0.080$

Q72. X and Y are independent with $\text{Var}(X) = 1$ and $\text{Var}(Y) = 2$. What is $\text{Var}(2X - 3Y)$?

(a) 8

(b) 22

(c) -4

(d) 14

Q73. Each impression is clicked independently with probability 0.02. The expected number of impressions up to and including the first click is:

(a) 2

(b) 20

(c) 50

(d) 98

Q74. The Central Limit Theorem states that, for i.i.d. samples with finite variance, the:

(a) Sample median equals the sample mean

(b) Sample variance converges to 0

(c) Distribution of the sample mean approaches a normal distribution as n grows, whatever the population's shape

(d) Population itself becomes normal as n grows

Q75. Treatment A has a higher success rate than B within every subgroup, but a lower rate overall. This is an example of:

(a) Simpson's paradox

(b) Regression to the mean

(c) The gambler's fallacy

(d) Survivorship bias

Q76. If $X \sim \mathcal{N}(0,1)$ and $Y = X^2$, then:

(a) X and Y are independent

(b) $\mathrm{Corr}(X,Y) = 0$, but X and Y are dependent

(c) $\mathrm{Corr}(X,Y) = 1$

(d) $\mathrm{Corr}(X,Y)$ is undefined

Bandits and Reinforcement Learning

Q77. With ε-greedy ($\varepsilon = 0.1$) over 5 arms, where exploration picks uniformly among all arms, what is the probability of pulling the current greedy arm?

(a) 0.10

(b) 0.98

(c) 0.90

(d) 0.92

Q78. UCB1 selects the arm that maximises:

(a) $\hat{\mu}_i$ only

(b) $\hat{\mu}_i + \sqrt{2 \ln t / n_i}$

(c) A random sample from the arm's posterior

(d) $\hat{\mu}_i - \sqrt{2 \ln t / n_i}$

Q79. Thompson sampling for Bernoulli-reward arms works by:

(a) Pulling a uniformly random arm with probability ε

(b) Always pulling the arm with the highest posterior mean

(c) Pulling each arm in round-robin order

(d) Sampling a mean from each arm's Beta posterior and pulling the arm with the highest sample

Q80. What distinguishes a contextual bandit from a full reinforcement-learning problem?

(a) In a contextual bandit, actions do not influence future contexts or states

(b) Full RL problems have only one state

(c) Contextual bandits cannot use features

(d) Contextual bandits have no reward signal

Q81. LinUCB assumes that:

(a) Rewards are always binary

(b) Each arm's expected reward is linear in the context feature vector

(c) The context is the same in every round

(d) The number of arms is at most two

Q82. In a multi-armed bandit, cumulative regret after T rounds is:

(a) The number of times the best arm was pulled

(b) The variance of the observed rewards

(c) The total reward collected

(d) The sum over rounds of the gap between the best arm's expected reward and the chosen arm's expected reward

Q83. The main weakness of inverse propensity scoring (IPS) for off-policy evaluation is:

(a) It is biased even with correct propensities

(b) It needs a model of the reward function

(c) High variance when the logging policy's propensities are small

(d) It works only for deterministic target policies

Q84. A logged impression shows action a chosen with logging propensity $\mu(a|x) = 0.2$ and reward 1. The target policy picks a with probability 0.6. This sample's IPS contribution is:

(a) 0.12

(b) 0.33

(c) 0.6

(d) 3.0

Q85. Q-learning differs from SARSA in that Q-learning:

(a) Requires a model of the transition dynamics

(b) Is on-policy: its target uses the next action actually taken

(c) Does not use a discount factor

(d) Is off-policy: its target uses $\max_{a'} Q(s', a')$ regardless of the action actually taken next

Q86. An agent receives reward 1 at every step forever, with discount factor $\gamma = 0.9$. The discounted return is:

(a) 9

(b) 10

(c) Infinite

(d) 0.9

Q87. In REINFORCE, subtracting a baseline $b(s)$ from the return:

(a) Changes the optimal policy

(b) Reduces the variance of the gradient estimate without introducing bias

(c) Introduces bias but reduces variance

(d) Increases variance but removes bias

Q88. PPO's clipped surrogate objective is designed to:

(a) Turn the problem into supervised learning on expert labels

(b) Prevent the policy from changing too much in a single update

(c) Make the value function exactly zero

(d) Remove the need for any exploration

Recommendation and Ads

Q89. Which approach best addresses the cold-start problem for a brand-new item with no interactions?

(a) Assign it the global average embedding permanently

(b) Drop the item until it gathers clicks organically

(c) Use pure collaborative filtering on the user–item matrix

(d) Use content features (text, image, category) and give the item some exploration traffic

Q90. Matrix factorisation with $k = 20$ latent dimensions for 1,000 users and 500 items (no biases) has how many parameters?

(a) 500,000

(b) 10,000

(c) 30,000

(d) 1,520

Q91. Two-tower (dual-encoder) models are popular for the retrieval stage mainly because:

(a) They model rich cross-features between user and item better than any other model

(b) They do not need training data

(c) They always give better ranking accuracy than cross-encoders

(d) Item embeddings can be precomputed and searched with approximate nearest-neighbour indexes

Q92. Users click top-ranked results more often regardless of relevance. A standard way to train a ranking model that corrects for this position bias is:

(a) Remove all clicks at position 1 from the training data

(b) Ignore it; position bias does not affect learned models

(c) Include position as a training feature (or weight by inverse examination propensity) and fix it to a constant at serving time

(d) Train only on the bottom positions

Q93. You train a CTR model keeping only a fraction w of the negatives (and all positives). To recover the true probability p from the model's output p', use:

(a) $p = \dfrac{p'}{p' + (1 - p')/w}$

(b) $p = p'$, since no correction is needed

(c) $p = w\,p'$

(d) $p = p'/w$

Q94. A CPC ad bids \$0.50 per click and its predicted CTR is 2%. Its eCPM (expected revenue per 1,000 impressions) is:

(a) \$25.00

(b) \$0.01

(c) \$1.00

(d) \$10.00

Python, Data Stack and Spark

Q95. In NumPy, adding an array of shape $(3, 1)$ to an array of shape $(4,)$ gives shape:

(a) $(3, 4)$

(b) An error, because the shapes are incompatible

(c) $(3, 1)$

(d) $(4, 3)$

Q96. In pandas, with `g = df.groupby('user')['clicks']`, what is the key difference between `g.transform('mean')` and `g.agg('mean')`?

(a) transform is always faster

(b) transform returns a result aligned to the original rows (same length); agg returns one row per group

(c) agg cannot compute means

(d) They return identical outputs

Q97. Why should preprocessing (scaling, imputation, encoding) be wrapped in a scikit-learn `Pipeline` when cross-validating?

(a) So preprocessing is fit only on each training fold, which prevents leakage into the validation fold

(b) Because Pipelines train models faster on GPUs

(c) To enable automatic feature selection

(d) Because estimators cannot be used outside a Pipeline

Q98. A DataFrame has index labels $[10, 20, 30]$. What do `df.iloc[0]` and `df.loc[0]` return?

(a) iloc[0] returns the row labelled 10; loc[0] raises a KeyError

(b) iloc[0] raises a KeyError; loc[0] returns the first row

(c) Both return the row labelled 10

(d) Both raise a KeyError

Q99. Given `a = np.arange(6).reshape(2, 3)`, what is `a.sum(axis=0)`?

(a) `[0, 1, 2]`

(b) `[3, 12]`

(c) `15`

(d) `[3, 5, 7]`

Q100. In Spark, why is `reduceByKey` generally preferred over `groupByKey` followed by a sum?

(a) groupByKey cannot be used on numeric values

(b) reduceByKey avoids the shuffle entirely

(c) groupByKey is a narrow transformation that cannot run in parallel

(d) reduceByKey combines values within each partition before the shuffle, so far less data moves across the network

Answer Key

Supervised Learning Fundamentals

1 **(d)** Averaging over more neighbours gives a smoother boundary: less sensitive to individual points (lower variance) but less flexible (higher bias).

2 **(a)** The ℓ_1 ball is a diamond whose vertices lie on the axes, so the optimum often sits where some coordinates are exactly zero. L1 is a Laplace prior; L2 is Gaussian.

3 **(a)** $-\log \mathcal{N}(w; 0, \tau^2 I) \propto \|w\|_2^2$, which is the ridge penalty.

4 **(c)** The negative log-likelihood of Bernoulli labels with $p = \sigma(w^\top x)$ is exactly the binary cross-entropy.

5 **(d)** The gap signals high variance (overfitting). Adding more features increases capacity and makes it worse; the others reduce variance.

6 **(c)** $X^\top X$ is invertible iff X has full column rank; collinearity or $d > n$ breaks that. Ridge's $+\lambda I$ fixes it.

7 **(c)** The dual solution has $\alpha_i > 0$ only for support vectors; deleting any other point leaves the boundary unchanged.

8 **(a)** Expanding the exponential as a Taylor series gives polynomial features of every degree, so the implicit feature map is infinite-dimensional.

9 **(b)** $P(x \mid y) = \prod_j P(x_j \mid y)$. Priors can be anything and the per-feature likelihoods need not be Gaussian.

10 **(c)** Generative models learn $P(x, y)$ (via $P(x \mid y)P(y)$); the others model $P(y \mid x)$ or a decision boundary directly.

11 **(b)** Tree splits are thresholds on one feature at a time, invariant to monotonic rescaling. Distance-based and penalised models are scale-sensitive.

12 **(c)** $\mathbb{E}[(y - \hat{f})^2] = (\mathbb{E}\hat{f} - f)^2 + \mathrm{Var}(\hat{f}) + \sigma^2$.

Decision Trees and Ensembles

13 **(a)** Gini $= 1 - \sum p_k^2 = 1 - (0.25 + 0.25) = 0.5$, the maximum for two classes.

14 **(d)** IG $= 1 - (0.5 \cdot 0 + 0.5 \cdot 0.918) = 0.541$.

15 **(d)** Averaging B trees with pairwise correlation ρ gives variance $\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$; feature subsampling lowers ρ. Fitting residuals is boosting.

16 **(a)** These pseudo-residuals equal ordinary residuals for squared loss. Reweighting points is AdaBoost; bootstrapping is bagging.

17 **(a)** Each tree contributes less, so more rounds are needed; small learning rate with many trees usually generalises better.

18 **(d)** $(1 - 1/n)^n \to e^{-1} \approx 0.368$. These out-of-bag points give a free validation estimate.

19 **(c)** Up-weighting mistakes forces the next weak learner to focus on the hard examples.

20 **(b)** It can fit the training data almost perfectly (low bias), but small data changes alter the tree a lot (high variance).

21 **(c)** XGBoost takes a second-order Taylor expansion of the loss; gain depends on $\sum g_i$ and $\sum h_i$ in each child plus λ, γ.

22 **(a)** Such features offer many candidate split points and can win by chance. Permutation importance on held-out data is less biased.

Unsupervised Learning and Clustering

23 **(a)** Lloyd's algorithm alternates assignment and centroid update; each step does not increase inertia, so it converges to a local optimum.

24 **(a)** k-means assigns each point to the nearest centroid, which gives convex Voronoi cells; it cannot represent non-convex shapes like rings.

25 **(b)** D^2 sampling spreads the seeds out and gives an $O(\log k)$-competitive expected cost.

26 **(b)** DBSCAN takes ε and minPts, not k. A single ε struggles when densities vary widely (HDBSCAN addresses this).

27 **(d)** $(6+3)/(6+3+1) = 0.9$.

28 **(b)** $s = (b-a)/\max(a,b)$ with a the mean intra-cluster distance and b the mean distance to the nearest other cluster. Near 0 means on a boundary.

29 **(a)** k-means is the limiting case of a GMM with shared spherical covariance $\sigma^2 I$ as $\sigma \to 0$ (hard assignments).

30 **(a)** Single linkage merges by the closest pair of points, so a bridge of points can join two clusters. Complete or Ward linkage give more compact clusters.

Neural Networks and Optimisation

31 **(b)** $\sigma'(z) = \sigma(z)(1-\sigma(z)) \le 0.25$, and it approaches 0 when the unit saturates.

32 **(a)** Often caused by a large learning rate or a large negative bias. Leaky ReLU or GELU keep a non-zero gradient for negative inputs.

33 **(d)** Using fixed running statistics makes the output for one example independent of the other examples in the batch.

34 **(c)** Scaling by $1/(1-p)$ during training keeps the expected activation unchanged, so inference needs no adjustment.

35 **(c)** Adam keeps exponential moving averages m_t and v_t, bias-corrects them, and updates with $\hat{m}_t/(\sqrt{\hat{v}_t}+\epsilon)$.

36 **(b)** ReLU zeroes half its inputs, so variance must double to preserve signal scale. Glorot was derived for tanh/linear units; zero init fails to break symmetry.

37 **(a)** $128 \times 64 + 64 = 8256$.

38 **(b)** $3 \cdot 3 \cdot 16 \cdot 32 + 32 = 4608 + 32 = 4640$.

39 **(d)** $\lfloor (32 + 2 \cdot 1 - 3)/2 \rfloor + 1 = 15 + 1 = 16$.

40 **(d)** The softmax Jacobian and the log combine to give the simple form $\partial L/\partial z = p - y$.

41 **(c)** Large steps overshoot the minimum and diverge. Lower the learning rate, add warm-up, or clip gradients.

42 **(a)** In Adam, an L2 gradient term gets divided by $\sqrt{\hat{v}}$, so weights with large gradients are decayed less. AdamW applies decay uniformly.

Transformers

43 **(c)** If q, k have unit-variance components, $q \cdot k$ has variance d_k; dividing by $\sqrt{d_k}$ restores unit variance.

44 **(b)** Every token attends to every other token: $QK^\top$ is $n \times n$, each entry costing $O(d)$. Projections add $O(nd^2)$.

45 **(b)** Setting future logits to $-\infty$ before the softmax keeps generation autoregressive. Random 15% masking is BERT's MLM.

46 **(c)** Each head has its own projections and can learn a different relation (syntax, coreference, position, and so on).

47 **(c)** Permuting the inputs permutes the outputs identically, so order must be injected (sinusoidal, learned, RoPE, ALiBi).

48 **(b)** BERT is a bidirectional encoder (MLM, plus next-sentence prediction originally). GPT-style decoders use causal next-token prediction.

49 **(b)** Each new token then needs only its own q, k, v and attention over cached K, V. This saves compute but uses memory linear in context length.

50 **(d)** Batch statistics are unstable with variable-length sequences and small batches; layer norm's per-token statistics avoid this.

51 **(d)** Four 512×512 matrices: $4 \times 262144 = 1048576$. Splitting into heads does not change the total.

52 **(d)** Sampling uses softmax(z/T): $T \to 0$ is greedy, $T > 1$ moves the distribution towards uniform.

Evaluation Methodology and Metrics

53 **(a)** Precision $= 40/50 = 0.8$, recall $= 40/60 = 0.667$, F1 $= 2PR/(P+R) = 2 \cdot 0.8 \cdot 0.667/1.467 \approx 0.727$.

54 **(b)** This ranking interpretation is why AUC is equivalent to the normalised Mann–Whitney U statistic.

55 **(b)** With extreme imbalance, a small FPR still means many false positives; ROC-AUC can look excellent while precision is poor. Accuracy is about 99.9% for a trivial model.

56 **(b)** The scaler's mean and variance then include test data, so test information reaches training. Put preprocessing inside a Pipeline.

57 **(a)** Random splits leak future patterns (trends, campaigns, user histories) into training and overstate offline performance.

58 **(b)** AUC depends only on the ranking of scores. The other three depend on the probability values themselves.

59 **(d)** DCG $= 1/\log_2 3 + 1/\log_2 4 = 0.631 + 0.5 = 1.131$. Ideal $[1,1,0]$: $1 + 0.631 = 1.631$. NDCG $= 1.131/1.631 \approx 0.69$.

60 **(d)** $(1 + 1/2 + 1/4)/3 = 1.75/3 \approx 0.583$.

61 **(d)** The inner loop selects hyperparameters; the outer loop estimates performance on data that the selection never saw.

62 **(b)** For keeping one user or time period inside a single fold, use GroupKFold or TimeSeriesSplit.

63 **(b)** Logged data only contains items the old policy showed, with position bias. Offline metrics are proxies; this is why counterfactual evaluation and A/B tests matter.

64 **(b)** The predictions are paired. McNemar's test uses only the discordant cases (one model right, the other wrong).

Probability and Statistics

65 **(c)** $\frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.0099}{0.0594} \approx 0.167$.

66 **(a)** It is computed under H_0; it is not the posterior probability of either hypothesis.

67 **(c)** SE $\propto 1/\sqrt{n}$, so $4\times$ the data gives $1/2$ the SE.

68 **(c)** Power $= 1 - \beta$. The second option is α (Type I error); the third is β (Type II error).

69 **(a)** $1 - 0.95^{20} \approx 0.64$. Corrections: Bonferroni (α/m) or Benjamini–Hochberg for FDR.

70 **(a)** The percentile bootstrap needs no normality assumption; it treats the empirical distribution as a stand-in for the population.

71 **(c)** Beta(1,1) + 7 successes, 93 failures gives Beta(8, 94), with mean $8/102$. The MLE would be 0.07.

72 **(b)** $\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}X + b^2 \mathrm{Var}Y$ for independent variables: $4 \cdot 1 + 9 \cdot 2 = 22$. Signs do not matter.

73 **(c)** The waiting time is geometric with mean $1/p = 50$.

74 **(c)** This underpins z-tests on CTR differences even though individual clicks are Bernoulli.

75 **(a)** It happens when subgroup sizes differ between treatments and a confounder links group membership with outcome.

76 **(b)** $\mathrm{Cov}(X, X^2) = \mathbb{E}[X^3] = 0$, yet Y is a function of X. Zero correlation rules out only linear dependence.

Bandits and Reinforcement Learning

77 **(d)** $0.9 + 0.1 \cdot \frac{1}{5} = 0.92$. The greedy arm can also be picked during exploration.

78 **(b)** Optimism in the face of uncertainty: rarely pulled arms get a larger bonus. Posterior sampling is Thompson sampling.

79 **(d)** Each arm is chosen with the posterior probability that it is optimal, which balances exploration and exploitation automatically.

80 **(a)** A bandit's reward is immediate and the next context is independent of the action, so there is no credit assignment over time.

81 **(b)** It keeps a ridge-regression estimate per arm and adds an upper-confidence bonus $\alpha\sqrt{x^\top A^{-1}x}$.

82 **(d)** Good algorithms (UCB, Thompson) have sub-linear regret, $O(\log T)$ problem-dependent; pure greedy can have linear regret.

83 **(c)** Weights $\pi(a|x)/\mu(a|x)$ explode when μ is small. Clipping, self-normalised IPS and doubly robust estimators reduce the variance.

84 **(d)** $r \cdot \pi(a|x)/\mu(a|x) = 1 \cdot 0.6/0.2 = 3$.

85 **(d)** Q-learning target: $r + \gamma \max_{a'} Q(s', a')$. SARSA target: $r + \gamma Q(s', a')$ with the a' actually taken. Both are model-free.

86 **(b)** Geometric series: $\sum_{t \geq 0} \gamma^t = 1/(1-\gamma) = 10$.

87 **(b)** $\mathbb{E}[\nabla \log \pi(a|s)\, b(s)] = 0$ because the score function has zero mean. A learned value function as baseline gives actor–critic.

88 **(b)** Clipping the ratio $\pi_\theta / \pi_{\text{old}}$ to $[1-\epsilon, 1+\epsilon]$ gives a cheap approximation to a trust region, as in TRPO.

Recommendation and Ads

89 **(d)** Pure CF has no signal for unseen items. Content-based or hybrid models combined with bandit exploration bootstrap them.

90 **(c)** $k(m+n) = 20 \cdot 1500 = 30000$, far fewer than the 500000 entries of the full matrix.

91 **(d)** Scoring is a dot product of independent towers, so retrieval over millions of items is fast. The price is no early user–item interaction, so a heavier ranker follows.

92 **(c)** Otherwise the model learns "was shown at the top" as a proxy for relevance and reinforces the old ranking.

93 **(a)** Downsampling multiplies the odds by $1/w$, so $\frac{p}{1-p} = w\frac{p'}{1-p'}$. Calibration matters because bids depend on p directly.

94 **(d)** eCPM $=$ bid $\times$ pCTR $\times$ 1000 $= 0.50 \times 0.02 \times 1000 = 10$. This is how CPC ads are ranked against CPM ads.

Python, Data Stack and Spark

95 **(a)** Broadcasting aligns trailing dimensions: $(3,1)$ and $(1,4)$ stretch to $(3,4)$.

96 **(b)** transform is handy for features such as "each row's user-level mean CTR".

97 **(a)** `cross_val_score(pipe, X, y)` refits the whole pipeline on each training fold.

98 **(a)** `iloc` is positional; `loc` is label-based, and there is no label 0.

99 **(d)** $a = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \end{bmatrix}$; axis 0 collapses the rows, giving column sums $[3, 5, 7]$.

100 **(d)** Both are wide transformations needing a shuffle, but map-side combining shrinks the data first. groupByKey ships every value and risks out-of-memory errors on skewed keys.